

Qingyun Yang

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Undergraduate researcher focused on solving *clinically meaningful* problems using *computer graphics, vision and machine learning*.

EDUCATION

Vanderbilt University B.S.

Aug 2022 - Present (expected May 2026)

Majors: Computer Science, Mathematics | Minors: Data Science

Nashville, Tennessee

GPA: 3.948/4 (Dean's List All Semesters)

RESEARCH EXPERIENCE

Progression-Aware Data Generation for Multi-Modality Brain Imaging

April 2025 – Present

Independent Researcher | *Generative Models, Learning-based Deformable Registration*

Johns Hopkins University (Remote)

- Proposed a generative framework that disentangles anatomical variance and pathology progression to simulate tumor progression in brain MRI from single timepoint scans, enabling longitudinal study under **Prof. Peirong Liu**.
- Currently building the core framework and preparing for model training using multi-modality brain imaging data.

3D Guidance via Nonrigid Registration for Accurate Tumor Re-Resection

January 2024 – Present

Independent Researcher | *Computer Vision, Graphics, Computer-assisted Interventions*

Vanderbilt University

- Led an independent research project on biomechanic-based deformable registration to correct tissue deformation for surgical visual guidance under **Prof. Jie Ying Wu**, improving alignment by **54%** over rigid registration.
- Developed a novel registration pipeline that utilizes both pre- and post-resection site surfaces as targets, improving accuracy by up to **33%** over previous methods
- Developed a script for promptable 3D point cloud segmentation and reconstruction from stereo imaging using the Segment Anything Model.
- Developed a script achieved automatic contour extraction from 3D meshes and point cloud using PCA projection.
- Developed a rigid registration algorithm combining Iterative Closest Point (ICP) and fiducial point alignment.
- Conducted **8** cadaver studies, collecting **23** En Bloc tissue specimens and reconstructing corrected 3D specimen models. Contributed to more accurate post-operative analysis of tissue deformation

Content Navigation in Body CT Images using Deep Regression Models

August 2024 – January 2025

Research Assistant | *Machine Learning, Medical Imaging*

Vanderbilt University

- Analyzed the generalizability of Body Part Regression technology across multi-contrast CT under **Prof. Bennett A. Landman**
- Pretrained and fine-tuned deep regression models on multi-contrast datasets, rigorously evaluating the influence of Global-Intensity-Nonlinear augmentation on model generalizability

PUBLICATION

- Yang, Q.***, Li, F.*, Xu, J., Liu, Z., Sridhar, S., Jin, W., ... & Wu, J. Y. (2025, September). Augmented Reality-Based Guidance with Deformable Registration in Head and Neck Tumor Resection. In International Conference on Medical Image Computing and Computer-Assisted Intervention (pp. 45-54). Cham: Springer Nature Switzerland. (**accepted paper, invited oral presentation, paper highlight**)
- Yang, Q.**, Acar, A., Ringel, M. J., Heiselman, J. S., Miga, M. I., Topf, M., & Wu, J. Y. (2025, April). Nonrigid alignment of en bloc tissue specimen to resection bed to enhance correspondence for re-resection guidance. In Medical Imaging 2025: Image-Guided Procedures, Robotic Interventions, and Modeling (Vol. 13408, pp. 238-246). SPIE. (**accepted paper, oral presentation**)
- H. Li, J. Wang, N. Kumar, T.E. Ertop, J. d'Almeida, D. Lu, A. Acar, J. Han, **Q. Yang**, J.Y. Wu, R.J. Webster III, I. Oguz. "Automated Segmentation of Central Airway Obstruction from Endoscopic Video Stream with Deep Learning." (2025, April). Automated segmentation of central airway obstruction from endoscopic video stream with deep learning. In Medical Imaging 2025: Image-Guided Procedures, Robotic Interventions, and Modeling (Vol. 13408, pp. 113-119). SPIE.
- D. Su, **Q. Yang**, K.D. Van Schaik, L.W. Remedios, Y. Zhang, T. Li, F. Maldonado, K.L. Sandler, B.M. Dawant, B.A. Landman. "BPPR: A Framework for Content Navigation in Multi-Contrast Body CT Images using Deep Regression Models." in IEEE Transactions on Biomedical Engineering. (under review)

TEACHING EXPERIENCE

Vanderbilt Computer Science department

Teaching Assistant

August 2023 - Present

Vanderbilt University

- Conducted 4 hours of weekly office hours, addressing inquiries from over 200 students across 3 sessions, enhancing the students' understanding of programming concepts
- Evaluated assignments and exams, offering detailed feedback on grammar, logic, and coding style, ensuring students develop strong programming skills and promoting best practices in code quality

HONORS & AWARDS

MICCAI 2025 NIH Registration Grant

- Selected by the NIH and MICCAI Society to support participation in the MICCAI 2025 conference.

2025 CRA Outstanding Undergraduate Researcher Award Honorable Mention

- Recognized by the Computing Research Association for excellence in undergraduate research in computing fields.

Stanford SERGE scholar

- Chosen for a highly selective Stanford Engineering program providing a fully funded PhD preparation experience.

WISE Fellowship 2025 (\$8000 funding)

WISE Fellowship 2024 (\$8000 funding)

WiC's Grace Hopper Celebration Scholarship

- Received full funding from Vanderbilt Women in Computing to attend Grace Hopper Celebration conference.

TECHNICAL SKILLS

Languages: Python, C++, Java, R, Racket, Prolog, MATLAB

Courses: Foundations of Machine Learning, Image Processing, Advanced Linear Algebra, Probability and Statistics for Engineering, Multi-variable Calculus, Methods of Ordinary Differential Equations, Intermediate Software Design, Data Structures, Algorithm, Computer Architecture, Electricity and Magnetism, Mechanics